

introduction applications, and highlights their relevance to the study of TDI systems and to a long standing conjecture of Paul Seymour on ideal clutters.

1.1 Totally dual in $\mathbb{L}$ systems

Consider a prime p . A rational number is p -adic if it is of the form $\frac{r}{s}$ where $r, s \in \mathbb{Z}$ and s is an integer power of p [2]. A number is $dyadic$ if it is 2-adic. Let $\mathcal{S}$ be a (possibly infinite) set of primes. We denote by $\mathbb{L}(\mathcal{S})$ the set of all rationals of the form $\frac{r}{s}$ where $r, s \in \mathbb{Z}$ and s is a product of integer powers of primes in $\mathcal{S}$. Observe that $\mathbb{L}(\{p\})$ denotes the p -adic rationals. To keep the notation light, we write $\mathbb{L}_p$ for $\mathbb{L}(\{p\})$. For any set of primes $\mathcal{S}$, $\mathbb{L}(\mathcal{S})$ is a heavy set (this follows from [3, Lemma 2.1]).

For a system $Mx \leq b$ to be TD in $\mathbb{L}$ we consider the dual $(D : M, b, w)$ for all choices of $w \in \mathbb{L}^n$. However, for the aforementioned heavy sets, it suffices to consider the choices $w \in \mathbb{Z}^n$. Namely,

Remark 1.1. *Let $\mathbb{L} := \mathbb{L}(\mathcal{S})$ where $\mathcal{S}$ is a set of primes. Then $Mx \leq b$ is TD in $\mathbb{L}$ if and only if for every admissible $w \in \mathbb{Z}^n$, $(D : M, b, w)$ has an optimal solution in $\mathbb{L}^m$.*

Proof. Necessity is clear, let us prove sufficiency. Consider an admissible $w \in \mathbb{L}^n$. For some μ that is a product of integer powers of primes in $\mathcal{S}$ we have $\mu w \in \mathbb{Z}^n$. There exists an optimal solution $\bar{y} \in \mathbb{L}^n$ for $(D : M, b, \mu w)$. However, then $\frac{1}{\mu}\bar{y}$ is a solution for $(D : M, b, w)$ that is in $\mathbb{L}^n$. $\square$

For any prime p , we can find a system $Mx \leq b$ that is TD in $\mathbb{L}_p$ but not TD in $\mathbb{L}_q$ for any prime $q \neq p$. Namely, one can pick $Mx \leq b$ to consist of a unique constraint, $px \leq 1$. However, if we require a system to be TD in $\mathbb{L}_p$ and TD in $\mathbb{L}_{p'}$ for distinct primes p and p' , then it is totally dual in $\mathbb{L}_q$ for every prime q [2, Theorem 1.4]. The following stronger statement (the aforementioned case corresponds to $\mathcal{S}_1 = \{p\}, \mathcal{S}_2 = \{p'\}$ and $k = 2$) holds in the full dimensional case,

Theorem 1.2. *Let $Mx \leq b$ be a system where $\{x : Mx \leq b\}$ is a full-dimensional polyhedron. For $i = 1, \dots, k$, let $\mathcal{S}_i$ be a set of primes and suppose that $Mx \leq b$ is TD in $\mathbb{L}(\mathcal{S}_i)$. If $\cap_{i \in [k]} \mathcal{S}_i = \emptyset$ then $Mx \leq b$ is TD in $\mathbb{L}_q$ for every prime q .*

This will be an immediate consequence of Theorem 2.2.

Consider an integral matrix M and an integral vector b . A necessary condition for $Mx \leq b$ to be TDI is that the polyhedron $Q = \{x \in \mathbb{R}^n : Mx \leq b\}$ be integral, i.e., that every minimal proper face of Q contains an integral vector [10], see also [20, Corollary 22.1a]. However, this is not a sufficient condition [20, Equation (3) in Chapter 22]. Let us define a stronger necessary condition for a system to be TDI. We say that $Mx \leq b$ is *near-TDI* if for every prime p , $Mx \leq b$ is TD in $\mathbb{L}_p$. Since $\mathbb{Z} \subset \mathbb{L}_p$ for every prime p , it then follows from Remark 1.1 that if a system is TDI, it is near-TDI.

Furthermore, we have the following result [3, Theorem 1.5],